Split Strategy and Deduplication Audit

What the Source Paper Pipeline Did Wrong

The main methodological problem in the source paper's released pipeline is that it does not cleanly separate retrieval data from evaluation data in the strict predictive process monitoring sense. In practice, the retrieval store is built in a way that allows prefixes coming from the same original log population to remain available when the test examples are later formed. This makes the setup closer to a retrieval-oriented experiment than to a strict held-out case-level benchmark.

A second problem is that deduplication is too weak. It is applied in a way that still allows effectively overlapping examples to survive across the retrieval side and the test side. As a result, the final evaluation can be helped by exact or near-exact repetitions that would not be acceptable in a stricter benchmark protocol.

A third problem is that the released implementation appears to define the label incorrectly in the prefix construction stage. Instead of always using the next unseen activity after the prefix, it can associate the example with the last activity already contained in the prefix itself. This creates a supervision inconsistency and weakens the reliability of the reported setup.

For these reasons, the released source-paper pipeline should not be treated as a clean benchmark reference for strict trace-level predictive process monitoring, even though the general research idea remains useful and relevant.

What the Current Thesis Pipeline Does Right Now

The current thesis pipeline is methodologically cleaner because it explicitly separates two different experimental settings instead of mixing them. The first is a row-level split, which is suitable for studying a retrieval-heavy setting where very similar historical prefixes may still be useful. The second is a trace-level split, which is the proper setting when the goal is to approximate case-level generalization.

In the current thesis pipeline, prefix construction and evaluation are internally consistent. The target of each example is the next activity after the observed prefix, and the retrieval collections are tied to the correct dataset variants. The naming of the vector collections is also structurally safe, so different preprocessed datasets are not accidentally mixed under the same identifier.

The present thesis pipeline is not yet the strictest literature-style protocol, because the trace split is currently random rather than chronological and because deduplication is not enforced globally across the final retrieval and test sides. However, these are limitations of evaluation strictness, not logical implementation errors. In other words, the current pipeline is coherent and interpretable, and it is already cleaner than the released source-paper implementation on the key leakage-related issues.

Therefore, the correct presentation is the following. The row-level split should be described as a retrieval-oriented experimental condition. The trace-level split should be described as the thesis pipeline's proper case-level evaluation mode. If an even stricter benchmark is desired later, the natural next step would be a chronological trace split aligned more closely with common predictive process monitoring practice [1], [2], [3].

Bibliography

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